

Camila Villalba Heuer

Backend Developer — Java — Spring Boot — Node.js — NestJS

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Professional Summary

Backend Developer with experience in **Java (Spring Boot)** and **Node.js (NestJS)**, building **REST APIs**, working with **SQL databases**, and implementing **JWT-based authentication and authorization** using Spring Security.

Background in **chemical and nuclear engineering**, with experience in **safety-critical environments**, bringing a strong focus on **robust system design, data consistency, validation, and failure prevention**. Experienced in designing **microservices architectures** using **Spring Cloud**.

Technical Skills

Backend: Java, Spring Boot, Node.js, NestJS, REST APIs, Spring MVC, JPA, Hibernate, Microservices, Spring Cloud

Security: Spring Security, JWT, OAuth2, role-based authorization, service-to-service authentication

Testing: JUnit, Mockito, MockMvc, Jest, Supertest

Databases: MySQL, PostgreSQL, relational modeling, joins, transactions, indexes, stored procedures

Tools: Docker, Git, GitHub, Maven, Swagger/OpenAPI

Other: React (basic), Python (Pandas, NumPy), Power BI

Professional Experience

Backend Developer — Java & Node.js Projects Aug 2025 – Present

- Designed and developed REST APIs using **Java (Spring Boot)** and **Node.js (NestJS)**
- Implemented **JWT authentication and role-based authorization** using Spring Security
- Modeled **MySQL databases**, ensuring data consistency, transactions, and integrity
- Built backend for a dental management system (patients, appointments, treatments)
- Developed unit and integration tests (JUnit, Mockito, Jest, Supertest)
- Used Docker for containerization and Git/GitHub for version control

Head of Maintenance & Operations – RA-10 Nuclear Reactor (CNEA) Mar 2023 – Oct 2025

- Led operations in a **nuclear research reactor**, ensuring safety and reliability
- Developed and optimized technical procedures and operational processes
- Performed technical analysis and troubleshooting in critical systems
- Coordinated cross-functional teams and continuous improvement initiatives

Selected Projects

Backend System – PsyCare (Psychological Practice Management API) (Java, Spring Boot, Spring Security, JWT, MySQL, Testing)

- Designed and developed a complete REST API for managing patients, appointments, clinical records, payments, and tutors
- Implemented **JWT-based authentication** and **stateless security** with Spring Security (custom filter, role-based access)
- Modeled complex domain logic including **minor patient–tutor relationships, appointment/payment state transitions**, and flexible patient registration flows
- Implemented dynamic appointment scheduling with availability calculation and business constraints
- Built multi-layered testing strategy: unit tests (JUnit, Mockito), controller tests (@WebMvcTest), and full **integration + security tests** using H2
- Designed error handling strategy with centralized exception management and consistent API responses
- Documented API with OpenAPI/Swagger and integrated CI pipeline with automated test execution (GitHub Actions)

link: [GitHub](#)

Microservices Architecture – Online Store (Java, Spring Boot, Spring Cloud, Docker)

- Designed microservices architecture using **Eureka, API Gateway, Config Server, OpenFeign**

- Implemented JWT-based security and **service-to-service authentication**
- Applied Resilience4j for fault tolerance
- Modeled explicit state transitions to ensure data consistency

link: **GitHub**

Education

Specialization in Technological Applications of Nuclear Energy

Balseiro Institute / UBA (2022)

Chemical Engineering (Environmental Orientation)

National University of Misiones (2015 – 2021)

Additional Training

Backend Development (Spring Boot, Microservices, Security, Testing)

SQL and Relational Databases

Python for Data Analysis (Pandas, NumPy, CI/CD)

Power BI and Data Analysis

Languages

Spanish (Native)

English (B1)